

"Physics as Information Processing" ~ Chris Fields ~

Lecture 5

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hello and welcome everyone it is September 14th 2023 and we are in the course physics as information processing it's lecture five looking forward to this lecture and then a little conversation Chris off to you thank you okay uh thank you Daniel uh and welcome to session five of physics of information processing uh and today we're going to talk about space time and um particularly about the idea that space-time is emergent from information processing and this has now become a a very mainstream idea in the quantum information and quantum gravity communities and they're still of course um some people holding out for the idea that space-time is fundamental but I would say that at least in these uh I expect it's become now the dominant idea that space-time is not fundamental that uh space-time is not part of the basic ontology of reality or something like that and that instead space-time is a construct so that's what today's session is going to be about so for those of you who are just joining or who would like a little review uh we started this course talking about Quantum information Theory and the idea that Quantum information theory is actually a new science about systems that communicate across a boundary and the systems are all always called Alice and Bob for kind of historical reasons and emphasize that this uh new Sciences topological it's about connectivity between systems it's not geometric so it doesn't assume a space-time embedding so it leaves open the possibility that space-time is actually a construct it doesn't assume space time up front and uh in the second session of the course I think we talked about how this boundary could always be represented as an array of qubits and that what the systems on each side of the boundary do to communicate is prepare and then measure the states of these qubits so they communicate by exchanging classical information via a Quantum Channel and this introduces a particular kind of quantum noise whenever they don't share of the reference frames that they use to to prepare the bits and measure the bits so it's it's not necessarily a perfect Communication channel from a classical Fidelity point of view but all of the noise is is quantum noise it's not classical and it always derives from differences in the reference frames

that are used by the two agents we reformulated the free energy principle uh in this setting and uh the free energy principle turns out to be very simple it turns out to be a classical limit of the principle of unitarity which is just the principle that information is conserved and we saw that the approach to uh perfect alignment between the two agents or or perfect prediction on the part of becomes more challenging not surprisingly a system complexity increases so the more complex the communication the harder it is to predict as one would expect uh in the last session we talked about how uh constraints on thermodynamic free energy induce compartmentalization in agents so whenever interact with their environments in a fairly complex ways then they end up being compartmentalized into compartments that have to communicate classically and this is solely a result of quantum information Theory you don't have to assume any biology to get this compartmentalization but clearly this is something that we see in biological systems not only do we see compartments we also also see hierarchical control uh which is likewise and a sort of automatic outcome of of the theory for complex systems and finally last time we talked a little bit about how these compartments communicate and introduce the idea of communication protocols that involve local operations on a shared Quantum resource of some kind and classical communication and we talked to Fairmount about how classical communication is a little bit problematic to Define in quantum theory because of course you're introducing classicality and so you have to make some very specific assumptions about what's classical and we'll see later on today uh what the key assumption to gives you classicality is uh as we talk about space-time but before uh I go into the topic of space-time I want to spend a little time um okay so this sorry this is this is where we're going we're talking about space time and we'll talk about applications to biology next time but before we get going on this I want to spend a little bit of time talking about explanation and this is motivated by a question that came up a very long question that touched on many different topics that came up in the interactive q a last time a fair amount of that question had to do with directly or indirectly but the question of what scale-free explanation looks like and how it compares to um scientific explanation in general most of which is reductive so I want to make a short detour into the question of how reductionist theories which all of you are probably very familiar with relate to scale-free theories uh and and particularly the the free energy principle is a scale-free theory so it's it's a theory that's unlike uh most of the theories that uh most of you probably learned about in college so we're all familiar with reductionist theories and the reductionist idea of scientific explanation which is is really an outcome of philosophy of science in the 20s and 30s people like Rudolph carnap were among the first to actually make

precise what this idea of reductive explanation is but the basic idea is that uh the laws or the Dynamics or the formalism that describes Behavior at some micro scale the behavior of little things actually explains the behavior of big things so the basic reduction of this idea is that uh what the little things are doing is what's important and what big things are doing uh are either emergent phenomena or they're just Epi phenomenal they're they're essentially just appearances and this defines a fundamental scale so the real assumption in any reductionist theory is that there's some fundamental scale at which the real dynamics takes place and everything else is just an appearance or it's some sort of emergent phenomenon and uh I try to be very careful when I use the word emergence because it's one of the most ambiguous words in the philosophy of science and people use it to mean all sorts of different things but whenever you say emergent uh it always sort of implies the idea that it's emergent from something and so the idea of emergence is essentially a reductionist idea uh because it at least implicitly assumes that there's some micro scale from which the macro scale is emergent and uh so the discussions of emergent phenomena are offered very consistent with reductionism even if they claim not to be we're all taught that science is reductionist and there's certainly people who who publicly claim that science is reduction is uh you know Richard Dawkins is certainly a shining example of this but I suspect that that a lot of scientists just kind of play lip service to reductionism Without Really believing in it uh because they don't really believe that the macroscopic phenomena that they are studying life for example is is completely determined by what's going on at the microscale or or what's going on at the micro scale plus some some stochastic noise or something like that so I want to explicitly contrast this with a scale-free theory in a scale-free theory is not simply a holistic Theory it's much more precise in a scale-free theory little things and big things obey the same laws or they have the same Dynamics or they're described by the same formalism so it's not that the macro scale is emergent from the micro scale the macro scale in the micro scale actually obey the same laws and the relationship between macro and micro is probably best stated as one of implementation macro scale things have micro scale components and those micro scale components are doing something or other and whatever they're doing implements the macro scale system and Its Behavior uh but this implementation of is not is not an explanation or at least it's not an explanation all by itself and the bumper sticker version of scale free theories is just as above so below whatever laws you have at the macro scale are the ones the same ones you have the micro scale and vice versa so scale free theories and a sensors simple it's the same Theory at every level of description so what is this idea of scale and how does it relate to

what we've talked about in this course well what we've talked about in this course all has to do with what an agent can observe on its boundary so scale has to do with how information is encoded on the boundary and we can describe that scale in many different ways uh here's a commonplace way to describe it in physics uh by a relationship between energy and distance or size you've probably all heard of the Planck scale uh the Planck scale is the smallest scale at which current physics makes any sense in energetic terms it's defined by energies on the order of 10^{19} eV which is giga electron volts the LHC the Large Hadron Collider in Switzerland it gets to about 10^4 eV so many many orders of magnitude less than the Planck scale so the point scale is extremely energetic compared to what we can probe experimentally the Planck scale energy of 10^{19} eV corresponds to a distance of about 10^{-35} meters which is incredibly tiny uh an intermediate scale on this diagram is about the scale of a nucleon so for example a proton or a neutron they have a rest mass of about one GeV and a size of about one femtometer so this intermediate scale is roughly the scale of nuclear physics and then if you go to our scale the scale of things that are you know about a meter in size of human beings tables and chairs lots of animals the things we're most familiar with that corresponds to an incredibly tiny energy 10^{-5} eV is about the energy of a photon of radio radiation and a meter is about the wavelength of long wave radio so if you think about visible light has a wavelength of about 10^{-6} meters an energy of about an electron volt so it's not surprising though that visible light feels energetic to us because it's orders of magnitude above our natural scale of this sort of thermal environment in which we exist so it's the encoding scale and the boundary that is going to relate theories of things that we see to each other and this ends up giving a kind of complicated picture if you think of Alice interacting with Bob and measuring things on the boundary sorry I'm a little bit ahead of myself I just wanted to show you a picture of a reductive Theory this is the Big Bang You're all familiar with this something happens at the Planck scale and the entire universe results from it by just a scale change so this is this is a reductive Theory a scale free theory is much more complicated so here's what a scale free Theory looks like uh Alice interacts with Bob and she's looking at her boundary and her boundary States the boundary State she sees could be encoded at different scales so there may be a small scale scale one and a bigger scale scale two and what Alice is looking for is some sort of theory that relates the first scale the micro scale to the macro scale so it's not that the micro scale came first like it did in the Big Bang and the macro scale came later uh these are these are encodings that can be happening simultaneously and the theoretical task is

to find theories that relate between these scales and these are implementation theories or you might call them embedding theories from one scale into another and what we want as a consistency Criterion is for these theories to be constant in time so as Bob evolves in time and so at Alice's are these encodings on her boundary she wants the relationship between encoding the different scales to stay constant because if it doesn't she'll never be able to figure anything out she'll just have this chaotic relationship between things happening at different scales and nothing will make so you have this complicated picture in scale-free theories and if you look at this and it seems Vaguely Familiar or even vaguely even very familiar I'll then uh it should because this is exactly the kind of explanatory structure or semantic structure that you have in computer science uh if you're studying the behavior of a computer which is just some some piece of Hardware some physical system then you can describe what the that piece of Hardware is doing at different scales and in fact what we call the hardware level of description when we talk about you know our laptops for example is typically the level of description of circuits of transistors inside microprocessors we think of that as the hardware level but of course that's just a description too that's layered on top of lots of much smaller scale Dynamics all the way down to the scale of of atoms and eventually nuclei and Elementary particles and all that other stuff so there's some Quantum stuff that's going on that we describe as having a classical Hardware level of bid Exchange and then we describe the behavior programs on top of that so what we want what makes a computer useful is that we can assign semantics to programs and we can talk about semantic relationships between programs for example the relationship between the internet uh protocol stuff that Zoom is doing and what you actually see on your user interface and here on your user interface that needs to stay fixed over time or Zoom becomes incomprehensible and and not programmable and in fact not even well defined as a as an algorithmic system and the same is true for the relationship between the operating system and the hardware description or between the operating system and and zoom as a program so computer science Works in exactly this scale-free way and so in a sense of scale free explanation is very familiar it's exactly the kind of explanation that we have in computer and that's why I refer to the relationship between scales is one of um because that's a term that we're familiar with from thinking about computers implementing programs so what are these theories I mean if we think in biological terms uh we can think of Pathways being embedded in a cell uh where cells being embedded in tissues all the way up to niches of various kinds which may be social niches being embedded in ecosystems and ecosystems being embedded in the biosphere and so we have all of these essentially semantic theories that talk about how

one scale embeds in another scale and if we have these sorts of of scale transitions we can think of representing the relationship between transitions at different scales in terms of the activity of some kind of renormalization group which is transforming entities and processes at one scale into entities and processes at another scale and I think in biology we can at least put forward the hypothesis in all the way from Pathways embedding in cells to ecosystems embedding in the biosphere we can put forward the hypothesis at least that the renormalization group flow across all of these scales is trivial that is that the embedding theories all have the same formal structure that of this the structure that describes how Pathways embed into cells is the same structure that describes how ecosystems embed into the biosphere now I certainly can't prove that uh I believe that it's likely so I think it's an interesting hypothesis to investigate and this is uh n hypothesis that I want to leave you with as a result of this class is something to think about oh to what extent and thinking about biological systems or even social systems are the embedding theories the same in thinking this way the physics of the situation is has the same formal structure as computer science has in terms of explanation oh we're of course faced with a problem which is that we're not just reconstructing physics we're talking about the physics of of interacting observers we're talking about the physics of communication so we're modeling How We Do physics and that makes our Theory self-referential and so we have to face the music from Godel's theorem which tells us essentially that powerful self-referential theories can't be both consistent and complete and by powerful it I just mean powerful enough to express arithmetic so not terribly powerful this becomes an issue that we have to live with we we will always be faced with this uh issue of having to construct meta theories which tell us how our lower level theories work but we never get to the end of that process we we never can get to theories that are both consistent and complete because our Tower of theories becomes progressively actually more powerful as it goes up so so Godel in a sense poses a problem but in another sense just tells us what life is like as a scientist oh that you can't have everything all the time doing science but we probably all already knew that so let's now talk about space uh end of digression and see how these ideas apply to a system or a tower of systems that can construct a spatial embedding in its World in its observed world so let's start out thinking about a that can see its environment but what it sees as a completely uniform environment where nothing is happening so if you have a uniform environment where nothing is happening uh you can't distinguish one part of the environment for another from another so so there's no you have no sense of space so what I want to do is build up from scratch what's needed to have a sense of space and then we'll we'll talk a little bit about what uh organisms

are doing and I think a very interesting question is if you look at phylogeny at what points in phylogeny what lineages in phylogeny actually need to have a sense of space and what do they have to have in terms of computational power to have that sense of space so if you just have a uniform environment uh you don't have a sense of space but the first the first thing we can do with an environment is to divide it up into some number of parts that are somehow distinguishable so suppose we can segment the environment into two different parts and if you think about E coli for example uh E coli has a bunch of sensors on the front and a bunch of effectors on the back so it can sense much more about the front part of the environment than the back part of the environment so already with E coli you have this kind of segmentation of the environment into parts but so far nothing is happening in this environment we just have segments that can be distinguished let's consider an environment like this that has distinguishable segments uh what one next wants to know about these segments is whether they're next to each other and if we just have a bunch of segments we just have a set and a set doesn't have any order or any relationship information so I've I've drawn this environment as a rectangle of rectangles but that's just a representation and it looks like there are relationships between these rectangles but if they're if they're just a set then there are no relationships if we want to add relationships then we have to add them explicitly so let's add some connections between these different segments of the environment how to turn that set into a graph that says explicitly for example that the light green segment is attached to the light blue segment and it's also attached to the dark green segment and the dark blue segment is attached to the light blue segment and attached to the dark green segment so now we have something that's that's much more mathematically complicated than a set we have a graph that has some connection information so we have a connection topology now adding this topology does something very important it breaks the exchange symmetry that is there if you just have a set so if we just have a set it doesn't make any difference if I exchange for example the positions of the upper right like blue segment with the upper left like green segment but if we have this connection topology then it does make a difference right I've I've uh Twisted the graph around and if I have to break connections to exchange things then I've actually changed the topology adding a topology adding connections can disrupt a symmetry that's there uh if there is no topology so now let's let's think about now we have the structure that's required to think about something happening so suppose that at some time is measured in the bit counter time reference frame of the observing system happens so a DOT appears in the light green segment and uh we can express this actually in the notation of quantum field Theory we can think of

[Music] the light green segment as a field and the dot appearing is the occurrence of an excitation in that field and so we can write this notation that says that the Green Field was excited to produce a DOT which is up Arrow Green Dot sometime after that oh it may turn out that the the dot disappears in the Green Field and some sort of dot appears down in the dark blue field and again we can write this in this kind of creation and destruction operator notation that that one has in Quantum field Theory and so this says that a Green Field is no longer excited but now the dark blue field is excited so this doesn't imply anything about these excitations being the same in fact they're completely different uh one is an excitation of the Green Field and one is an excitation of the blue field but at least now something is happening in the environment we can think about what does it take for an observer to notice that something is happening is in its environment it has to be able to to detect a change occurring in some segment of its environment and that change can involve something happening or something stopping happening in that segment of but it doesn't uh yet have to have any idea of what an object is but we can add that uh if the system has a memory we can have the system connect these events happening in different parts of its environment so if the system has a memory that has some number of steps I'll say in steps but here it could even just be two steps I'll it can see transitions between these two different states where there's a dot in the Green Field and then a dot in the in the dark blue field and then a dot in the Green Field and then a dot in the dark blue field and then a dot in the Green Field Etc so I've expressed this in this sort of field Theory notation over on the right side and what this is is a flip-flop it's an observed periodic Behavior so this is the basic idea behind a clock so any Observer that can see this some number of of time steps internally can interpret what's going on in its external environment as a clock ticking uh you can see some periodism as long as you can have memory and the ability to detect change you can see periodic behavior in the environment so you can see the environment acting like a clock so let's think a bit about what this does and does not require so seeing a clock in the environment requires having distinguishable non-exchangeable sectors of the environment so you have to have this connection topology that breaks the exchange symmetry it requires having these sectors Implement some sort of flip-flop so something periodic happening and you have to have enough memory to see that periodicity but it doesn't require any sense of object identity so the excitation in the green part and the excitation in the blue part can still just be field excitations of the green part of the blue part oh it doesn't there's there's nothing that says that those have to be the same thing they just have to appear and disappear in some periodic way and having a clock doesn't require a metric

space you don't need to have any sense of distance defined on the environment uh you just have to have this segmentation that is not rearrangeable so this tells us something extremely important which is that time is more fundamental than space I'm in the exterior environment is much simpler and easier to observe than space in the exterior environment so for thinking biologically about phylogeny we might expect to see organisms that can detect periodic behavior in the external environment long before they can detect spatial organization in the exterior environment and we know that organisms uh even down to to microbial mats can respond to the diurnal cycle we know that organisms in a sense can detect environmental clocks uh even organisms that we have no reason to believe divide the environment up into spaces or the sort of space that we're used to like a euclidean space so now about adding space to this idea of time in the environment the next step is to be able to see motion and being able to see one thing move from one segment of the environment to another segment as opposed to something happening in one segment and then something happens in another segment what has to have this idea of object persistence that there's a thing that has an identity and its identity doesn't change over time and it doesn't change when you move it someplace else and it appears someplace else in the environment it's the same object this is an idea that psychologists call object persistence and it's an idea that physicists call translational invariants translational invariance means two things it means that if I if I have an object and I move it to a different location I don't change the object it's the same object that I move from here to there so the location doesn't change the but it also means the reverse of that it means that the object doesn't change the location uh by by container remains invariant even though I move the object around in it so these are kind of the two faces of translational invariance and then the two phases of object persistence as the object moves through time its identity remains fixed and it doesn't change the time that it's moving through so if I'm measuring the moat it's motion with a clock uh moving the object doesn't change the clock these are clearly very classical ideas uh and we talked about earlier classical communication when we talked about classical Communication in for example the context of of a bell epr experiment we talked about The Observers being able to write down some data on a piece of paper or on a a disk drive or something and hand those data to someone else so this is translational invariance the data can't change when Alice gives it to Bob otherwise there isn't classical communication taking place and of course there's an up to noise built into here right the piece of paper might get wet and the ink might run a little bit but it's still the same message there's still some identity being assumed object persistence is not an idea we get from quantum theory it's an idea that we

impose on top of quantum theory and it's this idea that's intrinsically classical so we're looking right here at the quantum to classical transition this is it this is what gets added to Quantum Theory to get classical physics it's this idea of object persistence now once we have object persistence we can kind of go to town constructing here's how we do it we all we all learned about object persistence this way uh when we were six months old doing motor babbling and observing our hands and feeling how our bodies worked and seeing objects that we could manipulate and that didn't change when we manipulated them so this is the birth of object persistence this is the birth of classicality uh in in human psychology so now let's start thinking about building a coordinate system what's a coordinate system a coordinate system is just a set of labels that are attached to these segments of the environment and to build a coordinate system you have to have an origin and An Origin means an object something that's fixed something that doesn't move around and so a coordinate system requires the ability to identify a stigmergic memory requires the ability to find something in the environment that has a meaning that stays constant over time and that's that's what a stigmergic memory is it's something in the environment that provides us with a memory resource that we can count on being the same it's having this fixed point that allows us to label these connected segments of the how the contain that contain objects that move and be able to say okay the object moved from this part of the environment to another part and I can I can give those parts labels with respect to this other part that doesn't move this origin point and once I have this I can construct other invariances so for example I can consider an object that rotates around this origin without changing its identity and that defines what's called rotational invariance and I can also talk about an object that gets bigger and smaller compared to this fixed point and that defines radial distance and if you think about the the baby in the previous slide that's doing motor babbling it sees things getting smaller as it moves them farther away so the visual system does this computation for us and constructs this radial coordinate for us with respect to our bodies or specifically our heads our eyes so uh in in doing this this very early motor activity we're actually building the spatial coordinate system uh for our brains that's implemented by the hippocampus the place cells in the hippocampus and if I have an origin point in the uh I can also infer that I am moving that I have changed my location with respect to that object and that's a considerably more complicated inference because you have to have a representation of the self of you to say I moved so this is something that comes after you've developed this spatial coordinate system you have to be able to locate in a sense metacognitively a particular object that's you in that coordinate system so we're going to set that aside and now talk about

adding some numbers so adding a metric and adding a metric requires another particular kind of which is a ruler uh a particular fixed object that can be moved around and it has translational invariance and rotational invariance up to some transformation up to some tensor and in euclidean space that tensor is the identity so in a euclidean space you have perfect translational and rotational invariance as you get into general relativity then the ruler can actually change its length as you rotate or the ruler can change its length as it moves through so effectively your space time can deform the coordinates of your space time could deform and that gives you the representation of mass of course in general relativity so we talked uh several sessions ago about Quantum reference frames and I used the example of rulers and clocks as Quantum reference frames so now we see how to construct those uh from observations and symmetry breaking and object all of which we need to define a Quantum reference frame out in the environment like a ruler add notice that we don't need any of that environmental stuff to define those qrfs within our own processing and in fact uh the ones that exist within us we have no direct access to we can look at someone else's brain and talk about their spatial representation system but we can't look inside our own brains and talk about our spatial representation system so this is how we get a metric space uh from the ground up and we've talked about this from a very psychological or biological point of view starting from from sort of motor capabilities uh motor babbling for example uh a measure of effort which gives you some sense of mass so if it's easy to move something it's light and if it's hard to move something it's heavy and we understand that when we're six months old because we can measure how much force we're having to apply with our muscles we have this we have a reference internally for this force that we're applying and we use that internal measurement to attribute this idea of Mass to things in the environment we have to have memory for manipulations we have to be able to identify at variances that Define objects object persistence and we have to be able to build a ruler that we can move around and use to define a metric space it's very different from the way physicists approach this uh if you look at the physics literature then people start with information transfer they talk about uh entanglement as a basic resource there's a lot of discussion of bulk boundary Duality and the most well-known one is uh The Duality between AdS to Sitter space which is the sort of space you see inside a black hole and a conformal field Theory a Quantum field theory on the boundary of the black hole so things you can see for example going on on the horizon of a black hole or dual the geometry and the interior by this mathematical relationship you can use that mathematical relationship to construct a Quantum error correcting code by using the symmetries of the field

Theory as a resource for redundancy that allows you to represent the same thing in different places and as soon as you can do that you can build up a network of elements that look basically like elements of a lattice and a lattice uh is a graph to which you can assign symmetric relationships and and you get a metric space and in physics typically the goal is to get Einstein's equations out of this to start with some information theoretic sorts of constructs and get the equations of general relativity but what are the equations of general relativity I mean these are things that Einstein derived by doing fairly simple thought experiments using the invariances ordinary euclidean space combined with ideas from special relativity about the speed of light being constant and he added the the principle of equivalence which is the idea that essentially mass and acceleration are the same thing uh or the gravity and acceleration are the same the weight and acceleration of the same time so it's at least reasonable to expect this inferential process that's going on in theoretical physics is generating the same thing although in a very formalized way that we can generate by thinking about how organisms deal with their environments and so I want to leave this with you as the second hypothesis here one the first one was the hypothesis a trivial renormalization group flow across scales and biology the second hypothesis is that these two approaches to emergent space-time are actually giving us the same information and maybe starting out with the same information in fact that we may be able to think about these fundamental building blocks of information flow and entanglement in terms of the sorts of perceptual relationships that we've been talking in this class so there's our second hypothesis to think about going forward and I just want to leave you with the idea that organisms at least manage this very interesting process that we can think of starting as starting with with this the construct of memory organisms have memory multiple scales and it's memory that allows them to formulate the idea that there are objects and as soon as you have objects that persist you have a source of redundancy so you can do error correction and if you can do error correction if you can move the same thing from one place to another and have it encode the same information then you can build space talk and as soon as you have space time you can use the space time as more memory you can use the space time itself as an oh because space time has the nice ability to hold lots of copies of some informative structures so you can build libraries or you can build computers or you can build the internet if you have SpaceTime which gives you more memory which gives you more object persistence which gives you more error correction and the circle just spins around so organisms do this and the question is how does it get off the ground uh next time we can at least speculate about that kind of origin of Life type of question in the context of the fep and it's it's realization in simple systems of

communicating Observers so uh as always I encourage you to use the interactive q a uh the fifth discussion session with Andrew who's unfortunately not here today will be the last day of September uh Saturday at five as usual uh five o'clock European Time eight o'clock California time and then the last session of the class will be the 12th of October so uh thank you very much and I look forward to your questions in the interactive q a often thank you Chris um can I ask some short questions sure all right so first from Lana wrote is the periodicity the time lengths between and during the excitations built in so this was when we had the four cells and there was the inklings of a clock coming into play I think the simplest answer is that if nothing else is happening in the environment except this periodic Behavior then uh the it's not actually meaningful to talk yet about the time between the events so if if I have a if I have a periodic a system that's that's quasi-periodic for example and nothing else is going on in my then I can't distinguish that from a perfectly periodic system because I would have to have a perfectly periodic reference to know that the other system was aperiodic and if you think about the internal being driven by information flow into the system then something has to be happening in the environment for information to be flowing into the system and being counted so uh again as we talked about a few sessions ago the internal memory only gets updated when something happens so internal and external get tied together and I and this is this is why we can use external clocks but it's also why we depend on external clocks uh because our our intuitive sense of time is very malleable and so we we actually as humans use to correct for the malleability of our own uh intuitive sense of time wow and there's probably a lot to be said for that procrustean nature of having the chronometer the clock on the wall and the tick tick tick as an evolutionarily novel ruler of time playing a logistical role in time-bound Society yeah and you know think of think of earlier clocks that earlier cultures have used and that animals use I mean we have the the solar clock the lunar clock Etc and you know the the orbit of venous clock that was so important to the Mayans and so on so we we there are lots of available clocks in the environment uh if you have the ability to notice them and enough memory to realize that they're clocks awesome all right here's a question from upcycle Club they wrote how does representing the network structure and Dynamics with Quantum embeddings affect the semantic interpretation of the network the the embedding theory is the simplest semantic interpretation of the network and again a good way to think of this is through the lens of computer science you know as soon as you have a programming language that allows uh variable binding then you've built a very simple kind of semantics into the language um if you have some if you have X as a variable and you have variable binding then

at different times in the execution of the program uh X can have different causal consequences for Downstream computation because it's been bound to a different value so it has different semantics so the semantics of x again go back to go back to Gregory Bateson uh the the basis of semantics is differences that make a difference make a difference to behavior make a difference to actionability so uh variable binding gives you differences that actually make a difference to what the program does of so what is that is the semantics that's assigned by your embedding Theory um between what the programming language is doing and what the operating system is doing and so forth and I I think it's no surprise that the semantics of programming languages end up being represented with these very general mathematical constructs uh like category Theory which are essentially uh tools for representing arbitrary relations between systems and that's what embedding theories are there uh theories that state relationships between systems awesome and very provocative to explore on the biological and life and on the space time and matter sides which are of course seamlessly integrated through if only our own experiences yeah this this is another place where it's useful to go back to this difference between reductive theories and and scale free theories and scale free theories are all about semantics because they represent the existence of stuff happening at different scales and so they naturally raise the question of of what's what are these relationships uh what are these embedding theories in reductive theories uh there's a very deep sense in which semantics doesn't matter like there's if if you think that's that all that's going on that's important is the stuff that's going on at the Planck then you don't need any semantics right and so oh you you see this in late 60s you know early 70s uh old-fashioned AI where uh people talk about computation is purely syntactic and this is a very reductionist view of computation it leaves out the semantics and without the semantics the computation is useless it doesn't tell us anything at all unless we can map it to something we have this philosophy of science tradition that tells us that semantics is not important and we end up with tools like the fep that essentially tell us that semantics is all important and from a biological point of view you know the most natural response is of course we knew that all the time that's why we're doing biology so that all of this all of this ties would you say that fep says semantics is important and then what does it say about what semantics are important or is that entirely left up to the modeler well it tells you a lot about what's important of or the system itself right the system itself is dealing with and it's it's trying to increase its predictive power so it's building a theory of the and that theory is intrinsically semantic right it's about the and it's actionable right the the theory is meant to drive action on the that will either yield more information or

get the environment to to you know behaved to do what I'm predicting it's going to do uh I I think the fep has intrinsically semantic in that sense that it's about an agent modeling its environment yes and about the semantics of action as opposed to a partial or Limited semantics of passive inferors or semantics of abstraction which kind of leaves the the important link between the agent and the environment through sense making and decision selection basically unaddressed because it's kind of descartes you know in the thought experiment in the room and here when we actually engage with the boundary of the room that's where we start yeah yeah the fep ly destroys this this classical notion of of the passive Observer as as does quantum theory you there are no passive observers in quantum theory so this this is a construct that we inherited from a classical world view that is is seriously misleading a lot more to say but that's a great place to end thank you Chris okay thank you Daniel and and thanks to you all and and uh ask questions see you next time cheers